Two-Step Equity Underweight Strategy

Technical Model Documentation

Version 1.0 | CIO Quantitative Research

March 15, 2026

Abstract

This document provides a complete technical description of the Two-Step Equity Underweight Strategy. The model monitors four macro and market signals (payrolls momentum, yield-curve slope, HY credit spreads, and equity price trend) and dynamically reduces equity allocation from a neutral **50%** to a mild-underweight **35%** or severe-underweight **20%** when the composite signal falls below calibrated thresholds.

Over the full backtested period (December 1996 – November 2025, 29 years), the strategy delivered an annualised return of **8.14%** at **5.87%** annualised volatility, versus 7.97% at 15.38% for a fully invested equity position and 6.04% at 6.79% for a matched-exposure passive benchmark. The maximum drawdown was reduced from −52.6% (100% equities) to −10.7%, with an annualised Sharpe ratio of **1.39**.

Contents

1 Strategy Overview

The strategy is a rules-based, signal-driven overlay applied to a static mixed portfolio. The neutral allocation ($w_N = 50\%$ equity, 50% investment-grade bonds) is maintained unless the composite market signal crosses a calibrated threshold, at which point equity is reduced and the freed allocation is placed in USD cash.

1.1 Allocation States

Table 1: Allocation states and portfolio composition

State	Trigger Condition	Equity	Bonds	Cash
Neutral	$M_t > \theta_m$	50%	50%	0%
Mild UW	$\theta_s < M_t \leq \theta_m$	35%	50%	15%
Severe UW	$M_t \leq \theta_s$	20%	50%	30%

where M_t denotes the *Market Composite* signal (Section 3.6) and (θ_m, θ_s) are the mild and severe signal thresholds (Section 5). The bond allocation is held constant at 50% across all states. The de-risked equity portion earns the prevailing Federal Funds target rate.

1.2 Pipeline Architecture

The MATLAB implementation follows a strict Model–View–Controller (MVC) architecture using MATLAB packages:

- **`+Model/`** Signal generation, data access, backtesting logic.
- **`+View/`** GUI application (`UnderweightApp`).
- **`+Controller/`** Orchestration (`UnderweightController`).
- **`Underweight/`** Strategy scripts and helper functions.

The core execution pipeline is:

$$\text{buildSignals} \longrightarrow \text{labelEquityRegimes} \longrightarrow \text{thresholdApproach} \longrightarrow \text{evalPortfolio}$$

2 Data Sources

Table 2: Data sources used in the strategy

Series	Provider	Frequency	Available from
S&P 500 Index (`^SPX`)	Stooq	Daily OHLCV	January 1928
PAYEMS (Nonfarm Payrolls)	FRED	Monthly (SA)	January 1939
T10Y2Y (10Y−2Y Spread)	FRED	Daily (%)	June 1976
BAMLH0A0HYM2 (HY OAS)	FRED	Daily (%)	December 1996
FEDFUNDS (Effective FFR)	FRED	Monthly (%)	July 1954

All data is fetched via the `Model.DataManager` class, which maintains an on-disk SQLite registry and `.mat` cache. Incremental updates are performed automatically: a fresh request with no `startDate` only downloads observations newer than the last cached date.

The equity price priority chain is: Yahoo Finance (rate-limited) $\rightarrow$ Stooq SPY (data from 2005) $\rightarrow$ Stooq `^SPX` (data from 1994). The fallback to `^SPX` is triggered if the fetched history is shorter than 25 years (252×25 trading days).

The effective backtest start date is constrained by BAMLH0A0HYM2 (December 1996), though data is fetched from January 1994 to provide a 60-month warm-up for rolling z-score calculations.

3 Signal Construction

Each signal maps to the interval $(-1, +1)$ with the convention:

$$s_t > 0 \;\Leftrightarrow\; \textit{risk-on}, \qquad s_t < 0 \;\Leftrightarrow\; \textit{risk-off}.$$

Monthly signal values are obtained by taking the last non-NaN observation within each calendar month (end-of-month resampling).

3.1 Payrolls Momentum (`PayrollsMom`)

Source: FRED PAYEMS (Total Nonfarm Payrolls, thousands, monthly SA).

The signal is constructed via a five-step pipeline that provides a noise-robust, adaptive momentum indicator for the US labour market.

Step 1 — Raw month-over-month change

$$\Delta P_t = P_t - P_{t-1}$$

Step 2 — Trailing median prefilter (noise gate)

$$\widetilde{\Delta P}_t = \text{median}\big(\Delta P_{t-W+1}, \ldots, \Delta P_t\big), \quad W = 3$$

A single anomalous print (weather event, strike, seasonal revision) moves the trailing median of three to the middle value, effectively replacing the outlier with a neighbour. Two or more consecutive prints in the same direction pass through unchanged, preserving genuine labour-market inflections with at most one month of additional lag.

Step 3 — Exponential moving average (numerator path)

$$M_t = \alpha\, \widetilde{\Delta P}_t + (1-\alpha)\, M_{t-1}, \quad \alpha = 1 - e^{-\ln 2\,/\,h}, \quad h = 3 \text{ months}$$

Step 4 — Expanding-window winsorisation (denominator path)

$$Q_t = \text{quantile}\big(|\Delta P_1|, \ldots, |\Delta P_t|;\, p = 95\big), \qquad \widehat{\Delta P}_t = \text{sign}(\Delta P_t) \cdot \min\big(|\Delta P_t|, Q_t\big)$$

This prevents catastrophic single-month shocks (e.g. $-20{,}700$ K in April 2020 due to COVID-19) from inflating the rolling standard deviation and suppressing the signal for subsequent years.

Step 5 — Adaptive normalisation and tanh compression

$$\sigma_t = \text{std}\big(\widehat{\Delta P}_{t-N+1}, \ldots, \widehat{\Delta P}_t\big), \quad N = 60$$

$$\boxed{s_t^{\text{Pay}} = \tanh\left(\frac{M_t}{\sigma_t}\right) \in (-1, +1)}$$

Calibration: Under default parameters, a sustained expansion (+200 K/month) gives $s_t \approx +0.80$; a mild recession (-100 K/month) gives $s_t \approx -0.59$; an isolated shock is filtered to ≈ 0.

3.2 Yield Curve Slope (`YieldCurve`)

Source: FRED T10Y2Y (10-Year minus 2-Year Treasury Constant Maturity Rate, daily, percentage points).

Step 1 — Raw spread S_t = T10Y2Y level (percentage points). Positive = normal (risk-on); negative = inverted (risk-off).

Step 2 — Causal rolling z-score

$$z_t = \frac{S_t - \mu_t}{\sigma_t}, \quad \mu_t = \text{mean}(S_{t-W_d+1}, \ldots, S_t), \quad \sigma_t = \text{std}(S_{t-W_d+1}, \ldots, S_t)$$

where $W_d = W \times 21$ trading days, $W = 60$ months. A steep curve above its historical average yields $z > 0$; an inverted curve below its average yields $z < 0$.

Step 3 — EMA smoothing

$$m_t = \alpha\, z_t + (1-\alpha)\, m_{t-1}, \quad \alpha = 1 - e^{-\ln 2\,/\,h}, \quad h = 3 \text{ months}$$

Step 4 — Tanh compression

$$\boxed{s_t^{\text{YC}} = \tanh(m_t) \in (-1, +1)}$$

3.3 Credit Spread Signal (`CreditSpread`)

Source: FRED BAMLH0A0HYM2 (ICE BofA US High Yield Index Option-Adjusted Spread, daily, %).

The signal is *inverted*: wide spreads indicate credit stress and produce a negative (risk-off) output.

Step 1 — Raw OAS C_t = daily HY OAS level (%).

Step 2 — Expanding-window winsorisation (denominator path)

$$Q_t = \text{quantile}(C_1, \ldots, C_t;\ p = 97), \quad \hat{C}_t = \min(C_t, Q_t)$$

This caps the denominator series against extreme spike events (GFC 2008: $\approx 20\%$; COVID 2020: $\approx 11\%$) that would otherwise inflate σ_t and flatten the signal for years.

Step 3 — Causal rolling z-score

$$z_t = \frac{C_t - \mu_t(\hat{C})}{\sigma_t(\hat{C})}, \quad W = 60 \text{ months}$$

Step 4 — EMA smoothing

$$m_t = \alpha\, z_t + (1-\alpha)\, m_{t-1}, \quad \alpha = 1 - e^{-\ln 2\,/\,h}, \quad h = 1 \text{ month}$$

Step 5 — Tanh compression, inverted

$$\boxed{s_t^{\text{CS}} = \tanh(-m_t) \in (-1, +1)}$$

The short EMA half-life ($h = 1$) ensures the signal reacts immediately to credit market blowouts, which are typically fast-moving and highly predictive of equity drawdowns.

3.4 Equity Trend Signal (`EquityTrend`)

Source: S&P 500 daily close price P_t (Stooq `^SPX`).

Step 1 — Trailing moving average

$$\mathrm{MA}_t = \frac{1}{N}\sum_{i=0}^{N-1} P_{t-i}, \quad N = 200 \text{ days}$$

NaN is assigned for the first $N-1$ observations (warm-up period).

Step 2 — Tanh-compressed deviation

$$\boxed{s_t^{\mathrm{ET}} = \tanh\!\Big(k\left(\frac{P_t}{\mathrm{MA}_t} - 1\right)\Big) \in (-1, +1), \quad k = 10}$$

Calibration: 10% above the 200-day MA gives $\tanh(10 \times 0.10) = \tanh(1) \approx 0.76$; 10% below gives ≈ -0.76. The scale $k = 10$ provides a graded signal rather than a hard step function.

3.5 Equal-Weighted Composite (`Composite`)

$$\boxed{C_t = \frac{1}{n_t}\sum_{i\in\mathcal{S}} s_{i,t}, \quad n_t = \big|\{i \in \mathcal{S} : s_{i,t} \neq \mathrm{NaN}\}\big|}$$

where $\mathcal{S} = \{\text{PayrollsMom, YieldCurve, CreditSpread, EquityTrend}\}$. The NaN-tolerant averaging ensures the composite is defined even during the MA warm-up period and any data gaps.

3.6 Market Composite (`MarketComposite`)

A weighted composite that over-weights the two market-price signals (CreditSpread and EquityTrend) relative to the macro signal (YieldCurve):

$$\boxed{M_t = \frac{w_{\mathrm{CS}}\, s_t^{\mathrm{CS}} + w_{\mathrm{ET}}\, s_t^{\mathrm{ET}} + w_{\mathrm{YC}}\, s_t^{\mathrm{YC}}}{\sum_i w_i \cdot \mathbf{1}[s_{i,t} \neq \mathrm{NaN}]}}$$

Table 3: MarketComposite signal weights

Component	**Series**	**Weight**
Credit Spread	BAMLH0A0HYM2	30%
Equity Trend	S&P 500 vs 200d MA	50%
Yield Curve	T10Y2Y	20%
Total		**100%**

The denominator re-normalises to the sum of *available* weights, so the composite remains well-defined if any component is NaN. PayrollsMom is excluded from MarketComposite because its monthly frequency and lagged publication make it redundant relative to the higher-frequency market-price signals.

3.7 Monthly Resampling

All daily signals are resampled to a common monthly end-of-month grid:

$$s_{\text{month}} = \text{last non-NaN observation in calendar month}$$

The grid is anchored to the latest start month across all four series (December 1996, constrained by BAMLH0A0HYM2) and extends to the most recent complete month. A year$\times 100 +$month integer key is used for timezone-agnostic month matching.

4 Regime Classification

Forward-looking regime labels are assigned at each month t based on the subsequent three months of equity returns and realised volatility. These labels are used *only* for threshold calibration and decision-tree training; they are never used in live signal generation.

Let r^{eq}_{t+k} denote the equity log-return for month $t+k$. Define the 3-month forward cumulative return and annualised realised volatility:

$$R^{3m}_t = \prod_{k=1}^{3} (1 + r^{\text{eq}}_{t+k}) - 1, \qquad \sigma^{3m}_t = \text{std}(r^{\text{eq}}_{t+1}, r^{\text{eq}}_{t+2}, r^{\text{eq}}_{t+3}) \times \sqrt{12}$$

The label assigned to month t is:

$$\ell_t = \begin{cases} 2 & \text{(Severe UW)} \quad \text{if } R^{3m}_t < 0 \text{ and } \sigma^{3m}_t > 30\% \\ 1 & \text{(Mild UW)} \quad \text{if } R^{3m}_t < 0 \text{ and } \sigma^{3m}_t \leq 30\% \\ 0 & \text{(Neutral)} \quad \text{otherwise} \\ \text{NaN} & \text{if fewer than 3 forward months available} \end{cases}$$

The intuition is that high-volatility negative-return periods are the most destructive for long-term portfolio compounding and justify the most aggressive underweighting (Severe). Low-volatility negative-return periods (Mild) justify a moderate reduction.

4.1 Sample Distribution

Over the backtested period (December 1996 – November 2025, 348 labelled months):

Table 4: Regime label distribution

Regime	**Months**	**%**	**Allocation**
Neutral	242	69.5%	50% equity
Mild UW	87	25.0%	35% equity
Severe UW	19	5.5%	20% equity
Total	348	100%	

The severe class represents only 5.5% of observations, corresponding primarily to the GFC (2008–2009), COVID crash (2020), and the 2022 rate-hike bear market.

5 Signal Threshold Optimisation

5.1 Methodology

For each signal or composite s_t, the threshold pair $(\hat{\theta}_m, \hat{\theta}_s)$ is found by exhaustive grid search:

$$(\hat{\theta}_m, \hat{\theta}_s) = \arg\max_{\theta_s \le \theta_m} \mathrm{BalAcc}\big(a(s, \theta_m, \theta_s),\, \ell\big)$$

where the predicted allocation a_t maps to a label via:

$$\hat{\ell}_t(\theta_m, \theta_s) = \begin{cases} 2 & s_t \le \theta_s \\ 1 & \theta_s < s_t \le \theta_m \\ 0 & s_t > \theta_m \end{cases}$$

and balanced accuracy is the mean per-class recall:

$$\mathrm{BalAcc} = \frac{1}{3}\left(\frac{\mathrm{TP}_0}{|\ell = 0|} + \frac{\mathrm{TP}_1}{|\ell = 1|} + \frac{\mathrm{TP}_2}{|\ell = 2|}\right)$$

Balanced accuracy is used (rather than overall accuracy) to account for the severe class imbalance: a naïve classifier that always predicts Neutral achieves 69.5% overall accuracy but BalAcc of only 0.333.

The threshold grids used are:

$$\Theta_m \in \{-0.9, -0.8, \ldots, +0.9\}, \quad \Theta_s \in \{-0.9, -0.8, \ldots, +0.9\}$$

with the constraint $\theta_s \le \theta_m$, giving approximately 171 valid (θ_m, θ_s) pairs per signal.

5.2 Results

Table 5: Optimised thresholds and balanced accuracy by signal variant

Signal	**Mild thr $\hat{\theta}_m$**	**Severe thr $\hat{\theta}_s$**	**BalAcc**	**Avg Alloc**
PayrollsMom	−0.10	−0.30	0.498	45.9%
YieldCurve	+0.10	−0.80	0.261	35.4%
CreditSpread	−0.50	−0.60	0.469	41.2%
EquityTrend	+0.10	−0.40	0.510	43.1%
Composite	0.00	−0.20	0.499	43.9%
MacroComposite	0.00	−0.30	0.489	45.1%
MarketComposite	**−0.20**	**−0.40**	**0.492**	**44.1%**

The YieldCurve signal achieves the lowest balanced accuracy (0.261) because the yield curve is a slow-moving structural indicator; it leads equity drawdowns by 12–18 months on average, making near-term classification difficult. The EquityTrend signal achieves the highest individual BalAcc (0.510), reflecting the reflexive relationship between price momentum and regime.

The **MarketComposite** is selected as the primary signal for live allocation because it:

1. Combines both market-price signals (which lead on a 1–3 month horizon) with the macro yield signal (which provides leading information on recessions);
2. Achieves a competitive BalAcc of 0.492 at an average equity exposure of 44.1%;
3. Is robust to parameter perturbation given that its components react on different timescales.

6 Decision Tree Approach

As a secondary allocation method, a shallow classification tree is trained on all four signal features to predict the regime label. This approach allows non-linear interaction between signals without requiring explicit composite construction.

6.1 Model Specification

Algorithm CART (`fitctree`, Statistics and Machine Learning Toolbox).

Features $\mathbf{x}_t = [s_t^{\text{Pay}}, s_t^{\text{YC}}, s_t^{\text{CS}}, s_t^{\text{ET}}]^\top$.

Target Regime label $\ell_t \in \{0, 1, 2\}$.

Max splits 4 (constrains tree to at most 5 leaves for interpretability).

Cost matrix Asymmetric: misclassifying Severe as Neutral costs 5; misclassifying Mild as Neutral costs 2; symmetric otherwise. This penalises failure to de-risk during true stress regimes.

Validation 5-fold stratified cross-validation.

6.2 Trained Tree Rules

The tree trained on the full December 1996 – November 2025 sample yields the following rules:

1. If $s_t^{\text{Pay}} < -0.318$ and $s_t^{\text{ET}} < -0.396 \Rightarrow$ **Severe UW** (20% equity)
2. If $s_t^{\text{Pay}} < -0.318$ and $s_t^{\text{ET}} \geq -0.396 \Rightarrow$ **Neutral** (50% equity)
3. If $s_t^{\text{Pay}} \geq -0.318$ and $s_t^{\text{YC}} < -0.953 \Rightarrow$ **Mild UW** (35% equity)
4. If $s_t^{\text{Pay}} \geq -0.318$ and $s_t^{\text{YC}} \geq -0.953 \Rightarrow$ **Neutral** (50% equity)

Economic interpretation: Severe underweighting requires simultaneous deterioration in *both* labour markets (PayrollsMom below -0.318) *and* equity price momentum (EquityTrend below -0.396). Mild underweighting is triggered by a deeply inverted yield curve alone (YieldCurve below -0.953), consistent with yield-curve inversion as an intermediate-horizon recession signal.

6.3 Performance

Table 6: Decision tree classification performance

Metric	Value
In-sample balanced accuracy	0.552
5-fold CV balanced accuracy	0.391
Confusion matrix (rows = actual, cols = predicted):	
Neutral correctly classified	232 / 242
Mild correctly classified	10 / 87
Severe correctly classified	11 / 19

The CV accuracy of 0.391 vs. in-sample 0.552 indicates moderate overfitting, primarily from the under-represented Severe class (19 observations). The tree is offered as a supplemental tool; the *threshold approach* on MarketComposite is the primary allocation engine.

7 Portfolio Construction and Evaluation

7.1 Return Calculation

At each month t in the valid labelled subset, the strategy portfolio return is:

$$r_t^{\text{port}} = a_t\, r_t^{\text{eq}} + w_B\, r^{\text{bond}} + (w_B - a_t)\, r_t^{\text{cash}}$$

where:

- $a_t \in \{0.20, 0.35, 0.50\}$ = signal-implied equity allocation;
- $w_B = 0.50$ = constant bond weight;
- $r^{\text{bond}} = (1 + 0.04)^{1/12} - 1 \approx 0.327\%/\text{month}$ = assumed 4% p.a.;
- $r_t^{\text{cash}} = (1 + \text{FEDFUNDS}_t)^{1/12} - 1$ = monthly USD cash return from FRED FEDFUNDS (monthly average effective FFR).

Note that $w_B - a_t \geq 0$ for all valid states (since $a_t \leq w_B = 0.50$), so the total portfolio weight always sums to 1.00:

$$a_t + w_B + (w_B - a_t) = a_t + 0.50 + 0.50 - a_t = 1.00 \quad \checkmark$$

Portfolio returns at NaN (warm-up) months are treated as zero.

7.2 Benchmark Comparison

Three strategies are compared:

100% Equities $r_t = r_t^{\text{eq}}$. No risk management; full equity exposure throughout.

Matched Benchmark $r_t = \bar{a}\, r_t^{\text{eq}} + (1 - \bar{a})\, r^{\text{bond}}$, where $\bar{a} = 44.1\%$ is the time-average equity allocation of the Two-Step strategy. No timing; purely passive at the same long-run equity exposure.

Two-Step De-Risk The signal-driven strategy as described above.

The Matched Benchmark is the economically appropriate comparator because it eliminates the size effect: both strategies hold the same *average* equity position over time. Any excess return of the Two-Step strategy above the Matched Benchmark is attributable purely to the timing ability of the MarketComposite signal.

7.3 Performance Metrics

$$\hat{r}_{\text{ann}} = \left(\prod_{t=1}^{T} (1 + r_t) \right)^{12/T} - 1 \qquad \text{(annualised geometric return)} \qquad (1)$$

$$\hat{\sigma}_{\text{ann}} = \hat{\sigma}_m \sqrt{12} \qquad \text{(annualised volatility)} \qquad (2)$$

$$SR = \hat{r}_{\text{ann}} / \hat{\sigma}_{\text{ann}} \qquad \text{(annualised Sharpe ratio)} \qquad (3)$$

$$\text{MDD} = \min_t \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \qquad \text{(maximum peak-to-trough drawdown)} \qquad (4)$$

8 Empirical Results

8.1 Sample and Allocation Statistics

Table 7: Allocation state dwell statistics (December 1996 – November 2025)

State	**Months**	**%**	**Avg dwell**	**No. episodes**
Neutral (50%)	264	75.9%	20.3 months	13
Mild UW (35%)	32	9.2%	1.6 months	9
Severe UW (20%)	52	14.9%	3.5 months	11
Total	348	100%		

The strategy makes on average **1.62 allocation changes per year**, with the portfolio spending 75.9% of the time at the neutral 50% allocation. The average Neutral dwell of 20.3 months reflects the strategy's patience: it does not churn in and out at minor signal fluctuations.

8.2 Performance Summary

Table 8: 29-year backtest performance (December 1996 – November 2025)

Strategy	**Return p.a.**	**Vol p.a.**	**Max DD**	**Sharpe**	**Avg Eq**
100% Equities	7.97%	15.38%	−52.6%	0.52	100.0%
Matched Benchmark (44% eq)	6.04%	6.79%	−25.1%	0.89	44.1%
Two-Step De-Risk + Cash	**8.14%**	**5.87%**	**−10.7%**	**1.39**	**44.1%**

Bond rate assumption: 4.0% p.a. (constant) on the 50% bond allocation.
Cash rate: FRED FEDFUNDS; average over the sample period: **2.35% p.a.**

8.3 Key Observations

1. **Return enhancement:** The Two-Step strategy outperforms both 100% Equities (+17 bps p.a.) and the Matched Benchmark (+210 bps p.a.). The entire excess return over the Matched Benchmark is attributable to the timing skill of the MarketComposite signal, since both run at identical average equity exposure (44.1%).

2. **Volatility reduction:** Strategy volatility (5.87%) is 62% lower than 100% Equities (15.38%) and 14% lower than the already de-risked Matched Benchmark (6.79%). This arises from the strategy being underweight equity during high-volatility stress episodes.

3. **Drawdown control:** The maximum drawdown of −10.7% represents a reduction of 80% relative to 100% Equities (−52.6%) and 57% relative to the Matched Benchmark (−25.1%).

4. **Sharpe ratio:** The annualised Sharpe of **1.39** compares favourably with 0.89 (Matched Benchmark) and 0.52 (100% Equities), demonstrating that the timing signal generates positive risk-adjusted alpha.

5. **Structural bond allocation:** By holding a constant 50% in bonds, the strategy benefits from bond/equity diversification at all times, not just during risk-off episodes. The incremental cash position during underweighting additionally reduces mark-to-market risk in rising-rate environments.

9 Parameter Reference

Table 9: Complete model parameter reference

Parameter	Default	Description
Data parameters		
`EquityTicker`	SPY / `^SPX`	Equity ticker; fallback chain to `^SPX`
`StartDate`	1994-01-01	Data fetch start (BAMLH0A0HYM2 from 1996; pre-start used for warm-up)
Label thresholds		
`SevereRetThresh`	0	3m fwd return below this % qualifies for Severe
`SevereVolThresh`	0.30	3m fwd ann vol above this qualifies for Severe
`MildRetThresh`	0	3m fwd return below this % qualifies for Mild
Allocation weights		
`BaseWeight`	0.50	Neutral equity allocation
`MildWeight`	0.35	Mild underweight equity allocation
`SevereWeight`	0.20	Severe underweight equity allocation
EquityTrend signal		
`MAWindow`	200 (days)	Trailing price moving-average window
`TrendScale`	10	tanh scale k; 10 → 10% above MA ≈ 0.76
PayrollsMom signal		
`PrefilterWindow`	3 (months)	Trailing median noise-gate window
`SmoothHalfLife`	3 (months)	EMA half-life for numerator path
`NormWindow`	60 (months)	Rolling std window for denominator
`WinsorPct`	95	Expanding winsorisation percentile (denominator)
YieldCurve signal		
`ZScoreWindow`	60 (months)	Trailing z-score window
`SmoothHalfLife`	3 (months)	EMA half-life
`TanhScale`	1.0	tanh scale
CreditSpread signal		
`ZScoreWindow`	60 (months)	Trailing z-score window
`SmoothHalfLife`	1 (month)	EMA half-life (fast reaction to blowouts)
`TanhScale`	1.0	tanh scale (applied inverted)
`WinsorPct`	97	Expanding winsorisation percentile (denominator)
MarketComposite weights		
`w_CreditSpread`	0.30	Weight on CreditSpread component
`w_EquityTrend`	0.50	Weight on EquityTrend component
`w_YieldCurve`	0.20	Weight on YieldCurve component
Decision tree		
`TreeMaxSplits`	4	Maximum tree depth (→ 5 leaves)
`CostSevere`	5	Misclassification cost: Severe predicted as Neutral

(continued on next page)

Parameter	Default	Description
`CostMild`	2	Misclassification cost: Mild predicted as Neutral
Portfolio evaluation		
`BondAnnReturn`	0.04	Assumed constant annual bond return (50% allocation)
Cash rate	FEDFUNDS	FRED effective Federal Funds rate (monthly average)

Implementation Notes

1. **No-timezone datetimes:** Monthly grids use timezone-free MATLAB `datetime` to avoid `synchronize()` timezone conflicts between FRED, Stooq, and Yahoo data.
2. **Integer month-matching:** Year $\times 100 +$ month integer keys are used for all monthly alignment operations, providing unambiguous month matching across sources with different timestamp conventions.
3. **Cache system:** `Model.DataManager` maintains an SQLite registry (`cache/cache_registry.db`) and per-series `.mat` files. Incremental fetches avoid full re-downloads; a fresh request with explicit `startDate` bypasses the cache and fetches the full requested range.
4. **UIAxes compatibility:** All chart helpers use `ax` as the first argument and call `hold(ax,'on')` rather than `axes(ax)`, which is required for MATLAB App Designer `UIAxes` objects.
5. **Decision tree availability:** The decision tree approach requires the Statistics and Machine Learning Toolbox. If the toolbox is absent, `decisionTreeApproach` raises an error that the controller catches gracefully, continuing with threshold results only.